Question			Answer	Marks	AOs		Guidance
15	(a)		$\mathbf{W} = (-mg\sin 30°)\mathbf{i} + (-mg\cos 30°)\mathbf{j}$ $\left[\mathbf{W} = \left(-\frac{1}{2}mg\right)\mathbf{i} + \left(-\frac{\sqrt{3}}{2}mg\right)\mathbf{j}\right]$	**M1** **A1** **[2]**	**3.1b** **2.5**	Attempting to resolve the weight. Allow sin/cos interchange and sign errors for the method mark. All correct in this vector form	*mg* must be seen for the method mark.
15	(b)		$\mathbf{W} + \mathbf{R} + \mathbf{F} = \mathbf{0}$	**B1** **[1]**	**2.5**	Allow any rearrangement of this. Allow if their expression for **W** is used instead of **W**	
15	(c)		$\mathbf{R} = R\mathbf{j}$ $\left[(-mg\sin 30°)\mathbf{i} + (-mg\cos 30°)\mathbf{j} + (6\mathbf{i} + 8\mathbf{j}) + R\mathbf{j} = \mathbf{0}\right]$ **i** component $-mg\sin 30° + 6 = 0$ giving $m = 1.22$ to 3 sf **j** component: $-12\cos 30 + 8 + R = 0$ $R = 2.39$ so magnitude is 2.39 N	**B1** **M1** **A1** **A1** **M1** **A1** **[6]**	**3.1b** **3.1a** **1.1** **1.1** **3.1a** **3.2a**	Allow for any clear indication that **R** is a multiple of **j** or that it has no component in the **i** direction Forming equation from their **i** terms, or equivalent by resolving parallel to the plane. FT their **W** Correct equation in **i** direction **AG** Equation from the **j** terms (must include all three terms), oe, and using value of *m* Accept arwt 2.4	May be implied with an equation for the i direction with two terms and an equation in the j direction with three terms 12 is the value for *mg*. $1.22 \times 9.8 = 11.956$